JOINT ADMISSION TEST FOR MASTERS • IIT JAM MATHEMATICS (MA)

Differential Equations (2005–2026 Archive)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS	TOTAL MARKS	TEST DURATION	PAPER PATTERN
113	255 M	339 Min	MCQ • MSQ • NAT (2005–20

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	70	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	12	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	31	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

1. **Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
2. **Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
3. **Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
4. **Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
5. **Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
6. **Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at **vaibhavgeometer.github.io**.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q51 • ID: JAM_2026_Q51) NAT +2 M Neg: Nil (0)

Q.51	Let α, β and γ be fixed real numbers such that $y(x) = C_1 e^{-x} + C_2 e^{2x} + \alpha x e^{-x}$, where C_1 and C_2 are arbitrary real constants, is the general solution of the differential equation $$\frac{d^2y}{dx^2} + \beta \frac{dy}{dx} + \gamma y = -e^{-x}.$$ Then $\alpha(\beta + \gamma)$= _____(*rounded off to one decimal place*).

Question 2 (JAM 2026 • Q45 • ID: JAM_2026_Q45) NAT +1 M Neg: Nil (0)

Q.45	Let α and β be real numbers such that the differential equation $$(y^3 + \alpha x y^4 - 5x + \cos 2y)dx + (3xy^2 + 20x^2y^3 + \beta x \sin 2y)dy = 0$$ is exact. Then $\alpha + \beta$ = _____(*rounded off to one decimal place*).

Question 3 (JAM 2026 • Q35 • ID: JAM_2026_Q35) MSQ +2 M Neg: Nil (0)

Q.35	Consider the differential equation $$(2y\cos x - xy\ \sin x)\,dx + 2x\cos x\,dy = 0 \quad \text{for } x \in \left(0, \frac{\pi}{4}\right).$$ Which of the following is/are integrating factor(s) of the differential equation?
(A)	$\frac{1}{xy}$
(B)	xy
(C)	$\sec x$
(D)	$\sqrt{\sec x}$

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Question 4 (JAM 2026 • Q23 • ID: JAM_2026_Q23) MCQ +2 M Neg: −0.67

Q.23	Consider the differential equation $$\frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2y = \sin(e^x).$$ Which ONE of the following is a particular solution of this differential equation?
(A)	$-e^{-x}\sin(e^x)$
(B)	$-e^{-2x}\sin(e^x)$
(C)	$-e^{-x}(\sin(e^x) + \cos(e^x))$
(D)	$-e^{-2x}(\sin(e^x) + \cos(e^x))$

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Question 5 (JAM 2026 • Q18 • ID: JAM_2026_Q18) MCQ +2 M Neg: −0.67

Q.18	Let $\phi:\left(-\frac{\pi}{2},\frac{\pi}{2}\right)\to\mathbb{R}$ be the solution of the differential equation $$(\cos x)\frac{dy}{dx}-y=2y^2(\sin x-1)\cos x$$ satisfying $\phi(0)=\frac{1}{2}$. Then $\phi\left(\frac{\pi}{4}\right)=$_____.
(A)	2
(B)	$\frac{3+2\sqrt{2}}{2}$
(C)	$\frac{1}{2}$
(D)	$\frac{1}{\sqrt{2}}$

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Question 6 (JAM 2026 • Q16 • ID: JAM_2026_Q16) MCQ +2 M Neg: −0.67

Q.16	The orthogonal trajectories of the family of curves $y = -3x - 3 + me^{x}$ for a real parameter m are given by _____.
(A)	$x = \frac{y}{3} - \frac{1}{9} + k\, e^{-3y}$, k is a real parameter
(B)	$x = \frac{y}{3} - \frac{1}{9} + k\, e^{3y}$, k is a real parameter
(C)	$x = \frac{-y}{3} + \frac{1}{9} + k\, e^{-3y}$, k is a real parameter
(D)	$x = \frac{-y}{3} + \frac{1}{9} + k\, e^{3y}$, k is a real parameter

Question 7 (JAM 2026 • Q4 • ID: JAM_2026_Q4) MCQ +1 M Neg: −0.33

Q.4	The general solution of the differential equation $$x \sin\left(\frac{y}{x}\right)\frac{dy}{dx} = y \sin\left(\frac{y}{x}\right) + \frac{x}{2}, \quad x > 0$$ is given by _____.
(A)	$\cos\left(\frac{x}{y}\right) + \log_e(x^2) = K$, where K is an arbitrary constant
(B)	$\cos\left(\frac{y}{x}\right) + \log_e(x^2) = K$, where K is an arbitrary constant
(C)	$\cos\left(\frac{x}{y}\right) + \log_e(\sqrt{x}) = K$, where K is an arbitrary constant
(D)	$\cos\left(\frac{y}{x}\right) + \log_e(\sqrt{x}) = K$, where K is an arbitrary constant

Question 8 (JAM 2025 • Q56 • ID: JAM_2025_Q56) NAT +2 M Neg: Nil (0)

Q.56 Let $\varphi: \mathbb{R} \to \mathbb{R}$ be the solution of the differential equation

$$\frac{dy}{dx} + 2xy = 2 + 4x^2$$

satisfying $\varphi(0) = 0$.

Then, the value of $\varphi(2)$ is equal to _____ *(rounded off to two decimal places).*

Question 9 (JAM 2025 • Q55 • ID: JAM_2025_Q55) NAT +2 M Neg: Nil (0)

Q.55 Let $\varphi: (0, \infty) \to \mathbb{R}$ be the solution of the differential equation

$$x^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + y = 6x \ln x$$

satisfying $\varphi(1) = -3$ and $\varphi(e) = 0$.

Then, the value of $|\varphi'(1)|$ is equal to _____ *(rounded off to two decimal places).*

Question 10 (JAM 2025 • Q44 • ID: JAM_2025_Q44) NAT +1 M Neg: Nil (0)

Q.44 Let $\varphi: \mathbb{R} \to \mathbb{R}$ be the solution of the differential equation

$$4 \frac{d^2y}{dx^2} + 16 \frac{dy}{dx} + 25y = 0$$

satisfying $\varphi(0) = 1$ and $\varphi'(0) = -\frac{1}{2}$.

Then, the value of $\lim_{x \to \pi/6} e^{2x} \varphi(x)$ is equal to _____ *(rounded off to two decimal places).*

Question 11 (JAM 2025 • Q33 • ID: JAM_2025_Q33) MSQ +2 M Neg: Nil (0)

Q.33 Let $\varphi : (-1, \infty) \to (0, \infty)$ be the solution of the differential equation

$$\frac{dy}{dx} - 2ye^x = 2e^x\sqrt{y},$$

satisfying $\varphi(0) = 1$.

Then, which of the following is/are TRUE?

(A) φ is an unbounded function

(B) $\lim_{x \to \ln 2} \varphi(x) = (2e - 1)^2$

(C) $\lim_{x \to \ln 2} \varphi(x) = \sqrt{2e - 1}$

(D) φ is a strictly increasing function on the interval $(0, \infty)$

Question 12 (JAM 2025 • Q32 • ID: JAM_2025_Q32) MSQ +2 M Neg: Nil (0)

Q.32 If $M, N, \mu, w: \mathbb{R}^2 \to \mathbb{R}$ are differentiable functions with continuous partial derivatives, satisfying

$$\mu(x,y)M(x,y)dx + \mu(x,y)N(x,y)dy = dw$$

then which of the following is/are TRUE?

(A) μw is an integrating factor for $M(x,y)dx + N(x,y)dy = 0$

(B) μw^2 is an integrating factor for $M(x,y)dx + N(x,y)dy = 0$

(C) $w(x,y) = w(0,0) + \int_0^x (\mu M)(s,0)ds + \int_0^y (\mu N)(x,t)dt$, for all $(x,y) \in \mathbb{R}^2$

(D) $w(x,y) = w(0,0) + \int_0^x (\mu M)(s,y)ds + \int_0^x (\mu N)(0,t)dt$, for all $(x,y) \in \mathbb{R}^2$

Question 13 (JAM 2025 • Q29 • ID: JAM_2025_Q29) MCQ +2 M Neg: −0.67

Q.29 Let $\varphi : [0, \infty) \to \mathbb{R}$ be the continuous function satisfying

$$\varphi(x) = \left(\int_0^x \varphi(t)dt \right) + \sin x \ , \ \text{for all } x \in [0, \infty) .$$

Then, the value of $\lim_{x \to \pi/2} (2\varphi(x) - e^x)$ is equal to

(A) 1

(B) 2

(C) 3

(D) 4

Question 14 (JAM 2025 • Q14 • ID: JAM_2025_Q14) MCQ +2 M Neg: −0.67

Q.14 Let $\varphi : \mathbb{R} \to \mathbb{R}$ be the solution of the differential equation

$$\frac{dy}{dx} = (y-1)(y-3),$$

satisfying $\varphi(0) = 2$. Then, which one of the following is TRUE?

(A) $\lim_{x\to\infty} \varphi(x) = 0$

(B) $\lim_{x\to \ln\sqrt{2}} \varphi(x) = 1$

(C) $\lim_{x\to -\infty} \varphi(x) = 3$

(D) $\lim_{x\to \ln\frac{1}{\sqrt{2}}} \varphi(x) = 6$

Question 15 (JAM 2025 • Q12 • ID: JAM_2025_Q12) MCQ +2 M Neg: −0.67

Q.12 Let $\varphi : (0,\infty) \to \mathbb{R}$ be the solution of the differential equation

$$x\frac{dy}{dx} = (\ln y - \ln x)y,$$

satisfying $\varphi(1) = e^2$. Then, the value of $\varphi(2)$ is equal to

(A) e^2

(B) $2e^3$

(C) $3e^2$

(D) $6e^3$

Question 16 (JAM 2025 • Q11 • ID: JAM_2025_Q11) MCQ +2 M Neg: −0.67

Q.11 Let $\mathcal{C}$ denote the family of curves described by $yx^2 = \lambda$, for $\lambda \in (0, \infty)$ and lying in the first quadrant of the xy plane. Let $\mathcal{O}$ denote the family of orthogonal trajectories of $\mathcal{C}$.

Which one of the following curves is a member of $\mathcal{O}$, and passes through the point $(2, 1)$?

(A) $y = \frac{x^2}{4}, \ x > 0, \ y > 0$

(B) $x^2 - 2y^2 = 2, x > 0, \ y > 0$

(C) $x - y = 1, x > 0, \ y > 0$

(D) $2x - y^2 = 3, x > 0, \ y > 0$

Question 17 (JAM 2025 • Q5 • ID: JAM_2025_Q5) MCQ +1 M Neg: −0.33

Q.5 Which one of the following is the general solution of the differential equation

$$\frac{d^2y}{dx^2} - 8\frac{dy}{dx} + 16y = 2e^{4x} \ ?$$

(A) $\alpha_1 e^{4x} + \alpha_2 x e^{4x} + x^2 e^{4x}$, where $\alpha_1, \alpha_2 \in \mathbb{R}$

(B) $\alpha_1 e^{4x} + \alpha_2 x e^{4x} + 2x^2 e^{4x}$, where $\alpha_1, \alpha_2 \in \mathbb{R}$

(C) $\alpha_1 e^{-4x} + \alpha_2 e^{4x} + 2x^2 e^{4x}$, where $\alpha_1, \alpha_2 \in \mathbb{R}$

(D) $\alpha_1 x e^{-4x} + \alpha_2 x^2 e^{-4x} + x^2 e^{4x}$, where $\alpha_1, \alpha_2 \in \mathbb{R}$

Question 18 (JAM 2025 • Q2 • ID: JAM_2025_Q2) MCQ +1 M Neg: −0.33

Q.2 For which one of the following choices of $N(x, y)$, is the equation

$$(e^x \sin y - 2y \sin x)\, dx + N(x, y)\, dy = 0$$

an exact differential equation?

(A) $N(x, y) = e^x \sin y + 2 \cos x$

(B) $N(x, y) = e^x \cos y + 2 \cos x$

(C) $N(x, y) = e^x \cos y + 2 \sin x$

(D) $N(x, y) = e^x \sin y + 2 \sin x$

Question 19 (JAM 2024 • Q51 • ID: JAM_2024_Q51) NAT +2 M Neg: Nil (0)

Section C: Q.51 – Q.60 Carry TWO marks each.

Q.51 For $\alpha \in (-2\pi, 0)$, consider the differential equation

$$x^2\frac{d^2y}{dx^2} + \alpha x\frac{dy}{dx} + y = 0 \quad \text{for} \quad x > 0.$$

Let D be the set of all $\alpha \in (-2\pi, 0)$ for which all corresponding real solutions to the above differential equation approach zero as $x \to 0^+$. Then, the number of elements in $D \cap \mathbb{Z}$ equals _____

Question 20 (JAM 2024 • Q43 • ID: JAM_2024_Q43) NAT +1 M Neg: Nil (0)

Q.43 For $\alpha > 0$, let $y_\alpha(x)$ be the solution to the differential equation

$$2\frac{d^2y}{dx^2} - \frac{dy}{dx} - y = 0$$

satisfying the conditions

$$y(0) = 1, \quad y'(0) = \alpha.$$

Then, the smallest value of α for which $y_\alpha(x)$ has no critical points in $\mathbb{R}$ equals _____ (*rounded off to the nearest integer*).

Question 21 (JAM 2024 • Q42 • ID: JAM_2024_Q42) NAT +1 M Neg: Nil (0)

Q.42 Let $y: \mathbb{R} \to \mathbb{R}$ be the solution to the differential equation

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 5y = 1$$

satisfying $y(0) = 0$ and $y'(0) = 1$.

Then, $\lim_{x\to\infty} y(x)$ equals _____ (*rounded off to two decimal places*).

Question 22 (JAM 2024 • Q26 • ID: JAM_2024_Q26) MCQ +2 M Neg: −0.67

Q.26 Let $f: \mathbb{R} \to \mathbb{R}$ be a solution of the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 2e^x \text{ for } x \in \mathbb{R}.$$

Consider the following statements.

P: If $f(x) > 0$ for all $x \in \mathbb{R}$, then $f'(x) > 0$ for all $x \in \mathbb{R}$.

Q: If $f'(x) > 0$ for all $x \in \mathbb{R}$, then $f(x) > 0$ for all $x \in \mathbb{R}$.

Then, which one of the following holds?

(A) P is true but Q is false

(B) P is false but Q is true

(C) Both P and Q are true

(D) Both P and Q are false

Question 23 (JAM 2024 • Q23 • ID: JAM_2024_Q23) MCQ +2 M Neg: −0.67

Q.23 For $\alpha \in \mathbb{R}$, let $y_\alpha(x)$ be the solution of the differential equation

$$\frac{dy}{dx} + 2y = \frac{1}{1+x^2} \quad \text{for } x \in \mathbb{R}$$

satisfying $y(0) = \alpha$. Then, which one of the following is TRUE?

(A) $\lim_{x\to\infty} y_\alpha(x) = 0$ for every $\alpha \in \mathbb{R}$

(B) $\lim_{x\to\infty} y_\alpha(x) = 1$ for every $\alpha \in \mathbb{R}$

(C) There exists an $\alpha \in \mathbb{R}$ such that $\lim_{x\to\infty} y_\alpha(x)$ exists but its value is different from 0 and 1

(D) There is an $\alpha \in \mathbb{R}$ for which $\lim_{x\to\infty} y_\alpha(x)$ does not exist

Question 24 (JAM 2024 • Q18 • ID: JAM_2024_Q18) MCQ +2 M Neg: −0.67

Q.18 For the differential equation

$$y(8x - 9y)dx + 2x(x - 3y)dy = 0\,,$$

which one of the following statements is TRUE?

(A) The differential equation is not exact and has x^2 as an integrating factor

(B) The differential equation is exact and homogeneous

(C) The differential equation is not exact and does not have x^2 as an integrating factor

(D) The differential equation is not homogeneous and has x^2 as an integrating factor

Question 25 (JAM 2024 • Q10 • ID: JAM_2024_Q10) MCQ +1 M Neg: −0.33

Q.10 Let $g: \mathbb{R} \to \mathbb{R}$ be a continuous function. Which one of the following is the solution of the differential equation

$$\frac{d^2y}{dx^2} + y = g(x) \quad \text{for} \quad x \in \mathbb{R},$$

satisfying the conditions $y(0) = 0$, $y'(0) = 1$?

(A) $y(x) = \sin x - \int_0^x \sin(x-t)\, g(t)dt$

(B) $y(x) = \sin x + \int_0^x \sin(x-t)\, g(t)dt$

(C) $y(x) = \sin x - \int_0^x \cos(x-t)\, g(t)dt$

(D) $y(x) = \sin x + \int_0^x \cos(x-t)\, g(t)dt$

Madras

Question 26 (JAM 2024 • Q4 • ID: JAM_2024_Q4) MCQ +1 M Neg: −0.33

Q.4 Let $\mathcal{F}$ be the family of curves given by

$$x^2 + 2hxy + y^2 = 1, \quad -1 < h < 1\,.$$

Then, the differential equation for the family of orthogonal trajectories to $\mathcal{F}$ is

(A) $(x^2y - y^3 + y)\dfrac{dy}{dx} - (xy^2 - x^3 + x) = 0$

(B) $(x^2y - y^3 + y)\dfrac{dy}{dx} + (xy^2 - x^3 + x) = 0$

(C) $(x^2y + y^3 + y)\dfrac{dy}{dx} - (xy^2 + x^3 + x) = 0$

(D) $(x^2y + y^3 + y)\dfrac{dy}{dx} + (xy^2 + x^3 + x) = 0$

Madras

Question 27 (JAM 2024 • Q3 • ID: JAM_2024_Q3) MCQ +1 M Neg: −0.33

Q.3 Let $y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} = 1 + y \sec x \quad \text{for } x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$$

that satisfies $y(0) = 0$. Then, the value of $y\left(\frac{\pi}{6}\right)$ equals

(A) $\sqrt{3} \log\left(\frac{3}{2}\right)$

(B) $\left(\frac{\sqrt{3}}{2}\right) \log\left(\frac{3}{2}\right)$

(C) $\left(\frac{\sqrt{3}}{2}\right) \log 3$

(D) $\sqrt{3} \log 3$

Madras

Question 28 (JAM 2024 • Q1 • ID: JAM_2024_Q1) MCQ +1 M Neg: −0.33

Q.1 Let $y_c: \mathbb{R} \to (0, \infty)$ be the solution of the Bernoulli's equation

$$\frac{dy}{dx} - y + y^3 = 0, \qquad y(0) = c > 0.$$

Then, for every $c > 0$, which one of the following is true?

(A) $\lim_{x\to\infty} y_c(x) = 0$

(B) $\lim_{x\to\infty} y_c(x) = 1$

(C) $\lim_{x\to\infty} y_c(x) = e$

(D) $\lim_{x\to\infty} y_c(x)$ does not exist

Madras

Question 29 (JAM 2023 • Q59 • ID: JAM_2023_Q59) NAT +2 M Neg: Nil (0)

Q. 59 Let S be the set of all real numbers α such that the solution y of the initial value problem

$$\frac{dy}{dx} = y(2 - y),$$
$$y(0) = \alpha,$$

exists on $[0, \infty)$. Then the minimum of the set S is equal to _____.

(rounded off to two decimal places)

Question 30 (JAM 2023 • Q53 • ID: JAM_2023_Q53) NAT +2 M Neg: Nil (0)

Q. 53 Let $y : (1, \infty) \to \mathbb{R}$ be the solution of the differential equation

$$y'' - \frac{2y}{(1-x)^2} = 0$$

satisfying $y(2) = 1$ and $\lim_{x \to \infty} y(x) = 0$. Then $y(3)$ is equal to _____.
(rounded off to two decimal places)

Question 31 (JAM 2023 • Q43 • ID: JAM_2023_Q43) NAT +1 M Neg: Nil (0)

Q. 43 If y is the solution of

$$y'' - 2y' + y = e^x,$$
$$y(0) = 0, \quad y'(0) = -1/2,$$

then $y(1)$ is equal to _____. (rounded off to two decimal places)

Question 32 (JAM 2023 • Q33 • ID: JAM_2023_Q33) MSQ +2 M Neg: Nil (0)

Q. 33 Let $y : (\sqrt{2/3}, \infty) \to \mathbb{R}$ be the solution of

$$(2x - y)y' + (2y - x) = 0,$$
$$y(1) = 3.$$

Then

(A) $y(3) = 1$

(B) $y(2) = 4 + \sqrt{10}$

(C) y' is bounded on $(\sqrt{2/3}, 1)$

(D) y' is bounded on $(1, \infty)$

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Question 33 (JAM 2023 • Q22 • ID: JAM_2023_Q22) MCQ +2 M Neg: −0.67

Q. 22 Consider the family of curves $x^2 + y^2 = 2x + 4y + k$ with a real parameter $k > -5$. Then the orthogonal trajectory to this family of curves passing through $(2, 3)$ also passes through

(A) $(3, 4)$

(B) $(-1, 1)$

(C) $(1, 0)$

(D) $(3, 5)$

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Question 34 (JAM 2023 • Q4 • ID: JAM_2023_Q4) MCQ +1 M Neg: −0.33

Q. 4 Consider the initial value problem

$$\frac{dy}{dx} + \alpha y = 0,$$
$$y(0) = 1,$$

where $\alpha \in \mathbb{R}$. Then

(A) there is an α such that $y(1) = 0$

(B) there is a unique α such that $\lim_{x\to\infty} y(x) = 0$

(C) there is NO α such that $y(2) = 1$

(D) there is a unique α such that $y(1) = 2$

Question 35 (JAM 2022 • Q50 • ID: JAM_2022_Q50) NAT +1 M Neg: Nil (0)

Q.50 Consider the family $\mathcal{F}_1$ of curves lying in the region

$$\{(x, y) \in \mathbb{R}^2 : y > 0 \text{ and } 0 < x < \pi\}$$

and given by

$$y = \frac{c(1 - \cos x)}{\sin x}, \quad \text{where } c \text{ is a positive real number.}$$

Let $\mathcal{F}_2$ be the family of orthogonal trajectories to $\mathcal{F}_1$. Consider the curve $\mathcal{C}$ belonging to the family $\mathcal{F}_2$ passing through the point $\left(\frac{\pi}{3}, 1\right)$. If a is a real number such that $\left(\frac{\pi}{4}, a\right)$ lies on $\mathcal{C}$, then the value of a^4 is equal to _____.

(*Rounded off to two decimal places*)

Question 36 (JAM 2022 • Q49 • ID: JAM_2022_Q49) NAT +1 M Neg: Nil (0)

Q.49	Given that $y(x)$ is a solution of the differential equation $$x^2y'' + xy' - 4y = x^2$$ on the interval $(0, \infty)$ such that $\lim_{x\to 0^+} y(x)$ exists and $y(1) = 1$. The value of $y'(1)$ is equal to _____. (*Rounded off to two decimal places*)

Question 37 (JAM 2022 • Q47 • ID: JAM_2022_Q47) NAT +1 M Neg: Nil (0)

Q.47	Let $y(x)$ be the solution of the differential equation $$xy^2y' + y^3 = \frac{\sin x}{x} \quad \text{for } x > 0,$$ satisfying $y\left(\frac{\pi}{2}\right) = 0$. Then the value of $y\left(\frac{5\pi}{2}\right)$ is equal to _____. (*Rounded off to two decimal places*)

Question 38 (JAM 2022 • Q43 • ID: JAM_2022_Q43) NAT +1 M Neg: Nil (0)

Q.43	Let $y(x)$ be the solution of the differential equation $$\frac{dy}{dx} + 3x^2y = x^2, \quad \text{for } x \in \mathbb{R},$$ satisfying the initial condition $y(0) = 4$. Then $\lim_{x\to\infty} y(x)$ is equal to _____. (*Rounded off to two decimal places*)

Question 39 (JAM 2022 • Q35 • ID: JAM_2022_Q35) MSQ +2 M Neg: Nil (0)

Q.35	A real-valued function $y(x)$ defined on $\mathbb{R}$ is said to be **periodic** if there exists a real number $T > 0$ such that $y(x+T) = y(x)$ for all $x \in \mathbb{R}$. Consider the differential equation $$\frac{d^2y}{dx^2} + 4y = \sin(ax), \quad x \in \mathbb{R}, \qquad (*)$$ where $a \in \mathbb{R}$ is a constant. Then which of the following is/are true?
(A)	All solutions of $(*)$ are periodic for every choice of a.
(B)	All solutions of $(*)$ are periodic for every choice of $a \in \mathbb{R} - \{-2, 2\}$.
(C)	All solutions of $(*)$ are periodic for every choice of $a \in \mathbb{Q} - \{-2, 2\}$.
(D)	If $a \in \mathbb{R} - \mathbb{Q}$, then there is a unique periodic solution of $(*)$.

Question 40 (JAM 2022 • Q31 • ID: JAM_2022_Q31) MSQ +2 M Neg: Nil (0)

Q.31	Let $(-c, c)$ be the largest open interval in $\mathbb{R}$ (where c is either a positive real number or $c = \infty$) on which the solution $y(x)$ of the differential equation $$\frac{dy}{dx} = x^2 + y^2 + 1 \quad \text{with initial condition} \quad y(0) = 0$$ exists and is unique. Then which of the following is/are true?
(A)	$y(x)$ is an odd function on $(-c, c)$.
(B)	$y(x)$ is an even function on $(-c, c)$.
(C)	$\big(y(x)\big)^2$ has a local minimum at 0.
(D)	$\big(y(x)\big)^2$ has a local maximum at 0.

Question 41 (JAM 2022 • Q25 • ID: JAM_2022_Q25) MCQ +2 M Neg: −0.67

Q.25	For some real number c with $0 < c < 1$, let $\phi: (1-c, 1+c) \to (0, \infty)$ be a differentiable function such that $\phi(1) = 1$ and $y = \phi(x)$ is a solution of the differential equation $$(x^2 + y^2)dx - 4xy\,dy = 0.$$ Then which one of the following is true?
(A)	$\left(3\left(\phi(x)\right)^2 + x^2\right)^2 = 4x.$
(B)	$\left(3\left(\phi(x)\right)^2 - x^2\right)^2 = 4x.$
(C)	$\left(3\left(\phi(x)\right)^2 + x^2\right)^2 = 4\phi(x).$
(D)	$\left(3\left(\phi(x)\right)^2 - x^2\right)^2 = 4\phi(x).$

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Question 42 (JAM 2022 • Q23 • ID: JAM_2022_Q23) MCQ +2 M Neg: −0.67

Q.23	Consider the differential equation $$y'' + ay' + y = \sin x \quad \text{for } x \in \mathbb{R}. \qquad (**)$$ Then which one of the following is true?
(A)	If $a = 0$, then all the solutions of $(**)$ are unbounded over $\mathbb{R}$.
(B)	If $a = 1$, then all the solutions of $(**)$ are unbounded over $(0, \infty)$.
(C)	If $a = 1$, then all the solutions of $(**)$ tend to zero as $x \to \infty$.
(D)	If $a = 2$, then all the solutions of $(**)$ are bounded over $(-\infty, 0)$.

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Question 43 (JAM 2022 • Q21 • ID: JAM_2022_Q21) MCQ +2 M Neg: −0.67

Q.21	On the open interval $(-c, c)$, where c is a positive real number, $y(x)$ is an infinitely differentiable solution of the differential equation $$\frac{dy}{dx} = y^2 - 1 + \cos x,$$ with the initial condition $y(0) = 0$. Then which one of the following is correct?
(A)	$y(x)$ has a local maximum at the origin.
(B)	$y(x)$ has a local minimum at the origin.
(C)	$y(x)$ is strictly increasing on the open interval $(-\delta, \delta)$ for some positive real number δ.
(D)	$y(x)$ is strictly decreasing on the open interval $(-\delta, \delta)$ for some positive real number δ.

Question 44 (JAM 2022 • Q13 • ID: JAM_2022_Q13) MCQ +2 M Neg: −0.67

Q.13	Let $u: \mathbb{R} \to \mathbb{R}$ be a twice continuously differentiable function such that $u(0) > 0$ and $u'(0) > 0$. Suppose u satisfies $$u''(x) = \frac{u(x)}{1+x^2} \quad \text{for all } x \in \mathbb{R}.$$ Consider the following two statements: I. The function uu' is monotonically increasing on $[0, \infty)$. II. The function u is monotonically increasing on $[0, \infty)$. Then which one of the following is correct?
(A)	Both I and II are false.
(B)	Both I and II are true.
(C)	I is false, but II is true.
(D)	I is true, but II is false.

Question 45 (JAM 2021 • Q52 • ID: JAM_2021_Q52) NAT +2 M Neg: Nil (0)

Q. 52 The least possible value of k, accurate up to two decimal places, for which the following problem

$$y''(t) + 2y'(t) + ky(t) = 0, t \in \mathbb{R},$$
$$y(0) = 0, y(1) = 0, y(1/2) = 1,$$

has a solution is _____.

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Question 46 (JAM 2021 • Q48 • ID: JAM_2021_Q48) NAT +1 M Neg: Nil (0)

Q. 48 Let $y : \left(\frac{9}{10}, 3\right) \to \mathbb{R}$ be a differentiable function satisfying

$$(x - 2y)\frac{dy}{dx} + (2x + y) = 0, \quad x \in \left(\frac{9}{10}, 3\right), \quad \text{and } y(1) = 1.$$

Then $y(2)$ equals _____.

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Question 47 (JAM 2021 • Q25 • ID: JAM_2021_Q25) MCQ +2 M Neg: −0.67

JAM 2021 MATHEMATICS - MA

Q. 25 Let y be a twice differentiable function on $\mathbb{R}$ satisfying

$$y''(x) = 2 + e^{-|x|}, \quad x \in \mathbb{R},$$
$$y(0) = -1, \quad y'(0) = 0.$$

Then

(A) $y = 0$ has exactly one root.

(B) $y = 0$ has exactly two roots.

(C) $y = 0$ has more than two roots.

(D) there exists an $x_0 \in \mathbb{R}$ such that $y(x_0) \geq y(x)$ for all $x \in \mathbb{R}$.

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Question 48 (JAM 2021 • Q19 • ID: JAM_2021_Q19) MCQ +2 M Neg: −0.67

Q. 19 Consider the family of curves $x^2 - y^2 = ky$ with parameter $k \in \mathbb{R}$. The equation of the orthogonal trajectory to this family passing through $(1, 1)$ is given by

(A) $x^3 + 3xy^2 = 4$.

(B) $x^2 + 2xy = 3$.

(C) $y^2 + 2x^2y = 3$.

(D) $x^3 + 2xy^2 = 3$.

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Question 49 (JAM 2021 • Q12 • ID: JAM_2021_Q12) MCQ +2 M Neg: −0.67

Q. 12 Let y be the solution of

$$(1+x)y''(x)+y'(x)-\frac{1}{1+x}y(x)=0,\quad x\in(-1,\infty).$$
$$y(0)=1,\ y'(0)=0.$$

Then

(A) y is bounded on $(0,\infty)$.

(B) y is bounded on $(-1,0]$.

(C) $y(x)\geq 2$ on $(-1,\infty)$.

(D) y attains its minimum at $x=0$.

Question 50 (JAM 2021 • Q2 • ID: JAM_2021_Q2) MCQ +1 M Neg: −0.33

Q. 2 Let $P:\mathbb{R}\to\mathbb{R}$ be a continuous function such that $P(x)>0$ for all $x\in\mathbb{R}$. Let y be a twice differentiable function on $\mathbb{R}$ satisfying $y''(x)+P(x)y'(x)-y(x)=0$ for all $x\in\mathbb{R}$. Suppose that there exist two real numbers a,b $(a<b)$ such that $y(a)=y(b)=0$. Then

(A) $y(x)=0$ for all $x\in[a,b]$.

(B) $y(x)>0$ for all $x\in(a,b)$.

(C) $y(x)<0$ for all $x\in(a,b)$.

(D) $y(x)$ changes sign on (a,b).

Question 51 (JAM 2021 • Q1 • ID: JAM_2021_Q1) MCQ +1 M Neg: −0.33

Q. 1 Let $0<\alpha<1$ be a real number. The number of differentiable functions $y:[0,1]\to[0,\infty)$, having continuous derivative on $[0,1]$ and satisfying

$$\begin{aligned} y'(t) &= (y(t))^{\alpha},\quad t\in[0,1],\\ y(0) &= 0, \end{aligned}$$

is

(A) exactly one.

(B) exactly two.

(C) finite but more than two.

(D) infinite.

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Question 52 (JAM 2020 • Q57 • ID: JAM_2020_Q57) NAT +2 M Neg: Nil (0)

JAM 2020 MATHEMATICS - MA

Q. 57 If $x^2 + xy^2 = c$, where $c \in \mathbb{R}$, is the general solution of the exact differential equation

$$M(x, y)\, dx + 2xy\, dy = 0,$$

then $M(1, 1)$ is _____.

$\lceil 9 \quad 2 \quad 7 \quad 1 \rceil$

Question 53 (JAM 2020 • Q56 • ID: JAM_2020_Q56) NAT +2 M Neg: Nil (0)

Q. 56 Let $f : \mathbb{R} \to \mathbb{R}$ be a differentiable function with $f'(x) = f(x)$ for all x. Suppose that $f(\alpha x)$ and $f(\beta x)$ are two non-zero solutions of the differential equation

$$4\frac{d^2y}{dx^2} - p\frac{dy}{dx} + 3y = 0$$

satisfying

$$f(\alpha x)f(\beta x) = f(2x) \text{ and } f(\alpha x)f(-\beta x) = f(x).$$

Then, the value of p is _____.

Question 54 (JAM 2020 • Q48 • ID: JAM_2020_Q48) NAT +1 M Neg: Nil (0)

Q. 48 Let $f(x, y) = 0$ be a solution of the homogeneous differential equation

$$(2x + 5y)dx - (x + 3y)dy = 0.$$

If $f(x + \alpha, y - 3) = 0$ is a solution of the differential equation

$$(2x + 5y - 1)dx + (2 - x - 3y)dy = 0,$$

then the value of α is _____.

Question 55 (JAM 2020 • Q46 • ID: JAM_2020_Q46) NAT +1 M Neg: Nil (0)

JAM 2020 MATHEMATICS - MA

Q. 46 Consider the differential equation

$$\frac{dy}{dx} + 10y = f(x),\ x > 0,$$

where $f(x)$ is a continuous function such that $\lim_{x\to\infty} f(x) = 1$. Then the value of

$$\lim_{x\to\infty} y(x)$$

is _____.

Question 56 (JAM 2020 • Q35 • ID: JAM_2020_Q35) MSQ +2 M Neg: Nil (0)

Q. 35 Let $L[y] = x^2\frac{d^2y}{dx^2} + px\frac{dy}{dx} + qy$, where p, q are real constants. Let $y_1(x)$ and $y_2(x)$ be two solutions of $L[y] = 0, x > 0$, that satisfy $y_1(x_0) = 1, y_1'(x_0) = 0, y_2(x_0) = 0$ and $y_2'(x_0) = 1$ for some $x_0 > 0$. Then,

(A) $y_1(x)$ is not a constant multiple of $y_2(x)$

(B) $y_1(x)$ is a constant multiple of $y_2(x)$

(C) $1, \ln x$ are solutions of $L[y] = 0$ when $p = 1, q = 0$

(D) $x, \ln x$ are solutions of $L[y] = 0$ when $p + q \neq 0$

Question 57 (JAM 2020 • Q22 • ID: JAM_2020_Q22) MCQ +2 M Neg: −0.67

Q. 22 A solution of the differential equation $2x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} - y = 0, x > 0$ that passes through the point $(1, 1)$ is

(A) $y = \frac{1}{x}$ (B) $y = \frac{1}{x^2}$ (C) $y = \frac{1}{\sqrt{x}}$ (D) $y = \frac{1}{x^{3/2}}$

Question 58 (JAM 2020 • Q21 • ID: JAM_2020_Q21) MCQ +2 M Neg: −0.67

JAM 2020 MATHEMATICS - MA

Q. 21 Consider the differential equation $L[y] = (y - y^2)dx + xdy = 0$. The function $f(x, y)$ is said to be an integrating factor of the equation if $f(x, y)L[y] = 0$ becomes exact.

If $f(x, y) = \dfrac{1}{x^2y^2}$, then

(A) f is an integrating factor and $y = 1 - kxy,\ k \in \mathbb{R}$ is NOT its general solution

(B) f is an integrating factor and $y = -1 + kxy,\ k \in \mathbb{R}$ is its general solution

(C) f is an integrating factor and $y = -1 + kxy,\ k \in \mathbb{R}$ is NOT its general solution

(D) f is NOT an integrating factor and $y = 1 + kxy,\ k \in \mathbb{R}$ is its general solution

Question 59 (JAM 2020 • Q7 • ID: JAM_2020_Q7) MCQ +1 M Neg: −0.33

Q. 7 If $u = x^3$ and $v = y^2$ transform the differential equation $3x^5dx - y(y^2 - x^3)dy = 0$ to $\dfrac{dv}{du} = \dfrac{\alpha u}{2(u - v)}$, then α is

(A) 4 (B) 2 (C) -2 (D) -4

Question 60 (JAM 2019 • Q60 • ID: JAM_2019_Q60) NAT +2 M Neg: Nil (0)

Q.60 If $y(x)$ is the solution of the initial value problem

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0,\ y(0) = 2,\ \frac{dy}{dx}(0) = 0,$$

then $y(\ln 2)$ is (round off to 2 decimal places) equal to _____

END OF THE QUESTION PAPER

Question 61 (JAM 2019 • Q59 • ID: JAM_2019_Q59) NAT +2 M Neg: Nil (0)

Q.59 Let $y(x) = xv(x)$ be a solution of the differential equation

$$x^2\frac{d^2y}{dx^2} - 3x\frac{dy}{dx} + 3y = 0.$$

If $v(0) = 0$ and $v(1) = 1$, then $v(-2)$ is equal to_____

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Question 62 (JAM 2019 • Q21 • ID: JAM_2019_Q21) MCQ +2 M Neg: −0.67

Q.21 Let x, $x + e^x$ and $1 + x + e^x$ be solutions of a linear second order ordinary differential equation with constant coefficients. If $y(x)$ is the solution of the same equation satisfying $y(0) = 3$ and $y'(0) = 4$, then $y(1)$ is equal to

(A) $e + 1$ (B) $2e + 3$ (C) $3e + 2$ (D) $3e + 1$

Question 63 (JAM 2019 • Q20 • ID: JAM_2019_Q20) MCQ +2 M Neg: −0.67

Q.20 Let S be the family of orthogonal trajectories of the family of curves

$$2x^2 + y^2 = k, \text{ for } k \in \mathbb{R} \text{ and } k > 0.$$

If $\mathcal{C} \in S$ and $\mathcal{C}$ passes through the point $(1, 2)$, then $\mathcal{C}$ also passes through

(A) $(4, -\sqrt{2})$ (B) $(2, -4)$ (C) $(2, 2\sqrt{2})$ (D) $(4, 2\sqrt{2})$

Question 64 (JAM 2019 • Q7 • ID: JAM_2019_Q7) MCQ +1 M Neg: −0.33

Q.7 If $y(x) = \lambda e^{2x} + e^{\beta x}$, $\beta \neq 2$, is a solution of the differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$$

satisfying $\frac{dy}{dx}(0) = 5$, then $y(0)$ is equal to

(A) 1 (B) 4 (C) 5 (D) 9

Question 65 (JAM 2019 • Q6 • ID: JAM_2019_Q6) MCQ +1 M Neg: −0.33

Q.6 If $x^h y^k$ is an integrating factor of the differential equation

$$y(1+xy)\,dx + x(1-xy)\,dy = 0,$$

then the ordered pair (h,k) is equal to

(A) $(-2,-2)$ (B) $(-2,-1)$ (C) $(-1,-2)$ (D) $(-1,-1)$

Question 66 (JAM 2018 • Q60 • ID: JAM_2018_Q60) NAT +2 M Neg: Nil (0)

Q.60 If $y(x) = v(x)\sec x$ is the solution of $y'' - (2\tan x)\,y' + 5y = 0$, $-\frac{\pi}{2} < x < \frac{\pi}{2}$, satisfying $y(0) = 0$ and $y'(0) = \sqrt{6}$, then $v\left(\frac{\pi}{6\sqrt{6}}\right)$ is _____ (correct up to two decimal places).

END OF THE QUESTION PAPER

Question 67 (JAM 2018 • Q59 • ID: JAM_2018_Q59) NAT +2 M Neg: Nil (0)

Q.59 If $x(t)$ is the solution to the differential equation $\frac{dx}{dt} = x^2t^3 + xt$, for $t > 0$, satisfying $x(0) = 1$, then the value of $x\left(\sqrt{2}\right)$ is _____ (correct up to two decimal places).

Question 68 (JAM 2018 • Q46 • ID: JAM_2018_Q46) NAT +1 M Neg: Nil (0)

Q.46 Let $f:[0,\infty)\to[0,\infty)$ be continuous on $[0,\infty)$ and differentiable on $(0,\infty)$. If $f(x)=\int_0^x \sqrt{f(t)}\,dt$, then $f(6)=$ _____

Question 69 (JAM 2018 • Q32 • ID: JAM_2018_Q32) MSQ +2 M Neg: Nil (0)

Q.32 The solution(s) of the differential equation $\frac{dy}{dx}=(\sin 2x)\,y^{1/3}$ satisfying $y(0)=0$ is (are)

(A) $y(x)=0$

(B) $y(x)=-\sqrt{\frac{8}{27}}\,\sin^3 x$

(C) $y(x)=\sqrt{\frac{8}{27}}\,\sin^3 x$

(D) $y(x)=\sqrt{\frac{8}{27}}\,\cos^3 x$

Question 70 (JAM 2018 • Q24 • ID: JAM_2018_Q24) MCQ +2 M Neg: −0.67

Q.24 A particular integral of the differential equation $y''+3y'+2y=e^{e^x}$ is

(A) $e^{e^x}e^{-x}$ (B) $e^{e^x}e^{-2x}$ (C) $e^{e^x}e^{2x}$ (D) $e^{e^x}e^{x}$

Question 71 (JAM 2018 • Q23 • ID: JAM_2018_Q23) MCQ +2 M Neg: −0.67

Q.23 An integrating factor of the differential equation $\left(y+\frac{1}{3}y^3+\frac{1}{2}x^2\right)dx+\frac{1}{4}(x+xy^2)\,dy=0$ is

(A) x^2 (B) $3\log_e x$ (C) x^3 (D) $2\log_e x$

Question 72 (JAM 2018 • Q22 • ID: JAM_2018_Q22) MCQ +2 M Neg: −0.67

Q.22 Let $y(x)$ be the solution of the differential equation $\frac{dy}{dx} + y = f(x)$, for $x \geq 0$, $y(0) = 0$, where $f(x) = \begin{cases} 2, & 0 \leq x < 1 \\ 0, & x \geq 1 \end{cases}$. Then $y(x) =$

(A) $2(1 - e^{-x})$ when $0 \leq x < 1$ and $2(e - 1)e^{-x}$ when $x \geq 1$

(B) $2(1 - e^{-x})$ when $0 \leq x < 1$ and 0 when $x \geq 1$

(C) $2(1 - e^{-x})$ when $0 \leq x < 1$ and $2(1 - e^{-1})e^{-x}$ when $x \geq 1$

(D) $2(1 - e^{-x})$ when $0 \leq x < 1$ and $2\,e^{1-x}$ when $x \geq 1$

Question 73 (JAM 2018 • Q8 • ID: JAM_2018_Q8) MCQ +1 M Neg: −0.33

Q.8 In $\mathbb{R}^2$, the family of trajectories orthogonal to the family of asteroids $x^{2/3} + y^{2/3} = a^{2/3}$ is given by

(A) $x^{4/3} + y^{4/3} = c^{4/3}$

(B) $x^{4/3} - y^{4/3} = c^{4/3}$

(C) $x^{5/3} - y^{5/3} = c^{5/3}$

(D) $x^{2/3} - y^{2/3} = c^{2/3}$

Question 74 (JAM 2017 • Q60 • ID: JAM_2017_Q60) NAT +2 M Neg: Nil (0)

Q.60 If

$$y(x) = \int_{\sqrt{x}}^{x} \frac{e^t}{t}\,dt, \quad x > 0$$

Question 75 (JAM 2017 • Q51 • ID: JAM_2017_Q51) NAT +2 M Neg: Nil (0)

Q.51 Let $y(x),\ x > 0$ be the solution of the differential equation

$$x^2\frac{d^2y}{dx^2} + 5x\frac{dy}{dx} + 4y = 0$$

satisfying the conditions $y(1) = 1$ and $y'(1) = 0$. Then the value of $e^2y(e)$ is _____

Question 76 (JAM 2017 • Q41 • ID: JAM_2017_Q41) NAT +1 M Neg: Nil (0)

Q.41 If the orthogonal trajectories of the family of ellipses $x^2 + 2y^2 = c_1,\ c_1 > 0,$ are given by $y = c_2x^\alpha,\ c_2 \in \mathbb{R},$ then $\alpha =$ _____

Question 77 (JAM 2017 • Q40 • ID: JAM_2017_Q40) MSQ +2 M Neg: Nil (0)

Q.40 Let $k, \ell \in \mathbb{R}$ be such that every solution of

$$\frac{d^2y}{dx^2} + 2k\frac{dy}{dx} + \ell y = 0$$

satisfies $\lim_{x\to\infty} y(x) = 0$. Then

(A) $3k^2 + \ell < 0$ and $k > 0$

(B) $k^2 + \ell > 0$ and $k < 0$

(C) $k^2 - \ell \leq 0$ and $k > 0$

(D) $k^2 - \ell > 0, k > 0$ and $\ell > 0$

Question 78 (JAM 2017 • Q39 • ID: JAM_2017_Q39) MSQ +2 M Neg: Nil (0)

Q.39 Let $y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} = (y-1)(y-3)$$

satisfying the condition $y(0) = 2$. Then which of the following is/are TRUE?

(A) The function $y(x)$ is not bounded above

(B) The function $y(x)$ is bounded

(C) $\lim_{x\to+\infty} y(x) = 1$

(D) $\lim_{x\to-\infty} y(x) = 3$

Question 79 (JAM 2017 • Q30 • ID: JAM_2017_Q30) MCQ +2 M Neg: −0.67

Q.30 Let $y(x)$ be the solution of the differential equation

$$(xy + y + e^{-x})dx + (x + e^{-x})dy = 0$$

satisfying $y(0) = 1$. Then $y(-1)$ is equal to

(A) $\frac{e}{e-1}$ (B) $\frac{2e}{e-1}$ (C) $\frac{e}{1-e}$ (D) 0

Question 80 (JAM 2017 • Q29 • ID: JAM_2017_Q29) MCQ +2 M Neg: −0.67

Q.29 A particular integral of the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} = e^{2x}\sin x$$

is

(A) $\frac{e^{2x}}{10}(3\cos x - 2\sin x)$

(B) $-\frac{e^{2x}}{10}(3\cos x - 2\sin x)$

(C) $-\frac{e^{2x}}{5}(2\cos x + \sin x)$

(D) $\frac{e^{2x}}{5}(2\cos x - \sin x)$

Question 81 (JAM 2016 • Q56 • ID: JAM_2016_Q56) NAT +2 M Neg: Nil (0)

Q.56 Let $y(x)$ be the solution of the differential equation

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = 0, \qquad y(0) = 1, \qquad \left.\frac{dy}{dx}\right|_{x=0} = -1.$$

Then $y(x)$ attains its maximum value at $x =$ _____

Question 82 (JAM 2016 • Q47 • ID: JAM_2016_Q47) NAT +1 M Neg: Nil (0)

Q.47 Let $y_1(x)$, $y_2(x)$ and $y_3(x)$ be linearly independent solutions of the differential equation

$$\frac{d^3y}{dx^3} - 6\frac{d^2y}{dx^2} + 11\frac{dy}{dx} - 6y = 0.$$

If the Wronskian $W(y_1, y_2, y_3)$ is of the form ke^{bx} for some constant k, then the value of b is_____

Question 83 (JAM 2016 • Q35 • ID: JAM_2016_Q35) MSQ +2 M Neg: Nil (0)

Q.35 Consider the differential equation

$$\sin 2x \ \frac{dy}{dx} = 2y + 2\cos x \ , \ \ y\left(\frac{\pi}{4}\right) = 1 - \sqrt{2}.$$

Then which of the following statement(s) is(are) TRUE?

(A) The solution is unbounded when $x \to 0$
(B) The solution is unbounded when $x \to \frac{\pi}{2}$
(C) The solution is bounded when $x \to 0$
(D) The solution is bounded when $x \to \frac{\pi}{2}$

Question 84 (JAM 2016 • Q22 • ID: JAM_2016_Q22) MCQ +2 M Neg: −0.67

Q.22 The general solution of the differential equation with constant coefficients

$$\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

approaches zero as $x \to \infty$, if

(A) b is negative and c is positive
(B) b is positive and c is negative
(C) both b and c are positive
(D) both b and c are negative

Question 85 (JAM 2016 • Q21 • ID: JAM_2016_Q21) MCQ +2 M Neg: −0.67

Q.21 Let $y(x)$ be the solution of the differential equation

$$\frac{d}{dx}\left(x\frac{dy}{dx}\right) = x; \quad y(1) = 0, \quad \left.\frac{dy}{dx}\right|_{x=1} = 0 .$$

Then $y(2)$ is

(A) $\frac{3}{4} + \frac{1}{2}\ln 2$
(B) $\frac{3}{4} - \frac{1}{2}\ln 2$
(C) $\frac{3}{4} + \ln 2$
(D) $\frac{3}{4} - \ln 2$

Question 86 (JAM 2016 • Q9 • ID: JAM_2016_Q9) MCQ +1 M Neg: −0.33

Q.9 One of the points which lies on the solution curve of the differential equation

$$(y - x)dx + (x + y)dy = 0,$$

with the given condition $y(0) = 1$, is

(A) $(1, -2)$ (B) $(2, -1)$ (C) $(2, 1)$ (D) $(-1, 2)$

Question 87 (JAM 2016 • Q8 • ID: JAM_2016_Q8) MCQ +1 M Neg: −0.33

Q.8 The nonzero value of n for which the differential equation

$$(3xy^2 + n^2x^2y)dx + (nx^3 + 3x^2y)\,dy = 0, \quad x \neq 0,$$

becomes exact is

(A) -3 (B) -2 (C) 2 (D) 3

Question 88 (JAM 2015 • Q36 • ID: JAM_2015_Q36) MSQ +2 M Neg: Nil (0)

Q.6 The initial value problem

$$y' = \sqrt{y}, \; y(0) = \alpha, \; \alpha \geq 0$$

has

(A) at least two solutions if $\alpha = 0$

(B) no solution if $\alpha > 0$

(C) at least one solution if $\alpha > 0$

(D) a unique solution if $\alpha = 0$

Question 89 (JAM 2015 • Q24 • ID: JAM_2015_Q24) MCQ +2 M Neg: −0.67

Q.24 If $y(t)$ is a solution of the differential equation $y'' + 4y = 2e^t$, then

$$\lim_{t\to\infty} e^{-t}\, y(t)$$

is equal to

(A) $\frac{2}{3}$ (B) $\frac{2}{5}$ (C) $\frac{2}{7}$ (D) $\frac{2}{9}$

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Question 90 (JAM 2015 • Q12 • ID: JAM_2015_Q12) MCQ +2 M Neg: −0.67

Q.12 The orthogonal trajectories of the family of curves $y = C_1 x^3$ are

(A) $2x^2 + 3y^2 = C_2$ (B) $3x^2 + y^2 = C_2$
(C) $3x^2 + 2y^2 = C_2$ (D) $x^2 + 3y^2 = C_2$

Question 91 (JAM 2015 • Q9 • ID: JAM_2015_Q9) MCQ +1 M Neg: −0.33

Q.9 An integrating factor of the differential equation

$$\frac{dy}{dx} = \frac{2xy^2 + y}{x - 2y^3}$$

is

(A) $\frac{1}{y}$ (B) $\frac{1}{y^2}$ (C) y (D) y^2

Question 92 (JAM 2015 • Q8 • ID: JAM_2015_Q8) MCQ +1 M Neg: −0.33

Q.8 If an integral curve of the differential equation $(y - x)\frac{dy}{dx} = 1$ passes through $(0, 0)$ and $(\alpha, 1)$, then α is equal to

(A) $2 - e^{-1}$ (B) $1 - e^{-1}$ (C) e^{-1} (D) $1 + e$

Question 93 (JAM 2015 • Q2 • ID: JAM_2015_Q2) MCQ +1 M Neg: −0.33

Q.2 Let $y(x) = u(x)\sin x + v(x)\cos x$ be a solution of the differential equation $y'' + y = \sec x$. Then $u(x)$ is

(A) $\ln|\cos x| + C$
(B) $-x + C$
(C) $x + C$
(D) $\ln|\sec x| + C$

Question 94 (JAM 2014 • Q16 • ID: JAM_2014_Q16) MCQ +2 M Neg: −0.67

Q.16 The value of $\alpha \in \mathbb{R}$ for which the curves $x^2 + \alpha y^2 = 1$ and $y = x^2$ intersect orthogonally is

(A) -2 (B) $\frac{-1}{2}$ (C) $\frac{1}{2}$ (D) 2

Question 95 (JAM 2014 • Q15 • ID: JAM_2014_Q15) MCQ +2 M Neg: −0.67

Q.15 The equation of the curve passing through the point $\left(\frac{\pi}{2}, 1\right)$ and having slope $\frac{\sin(x)}{x^2} - \frac{2y}{x}$ at each point (x, y) with $x \neq 0$ is

(A) $-x^2y + \cos(x) = \frac{-\pi^2}{4}$
(B) $x^2y + \cos(x) = \frac{\pi^2}{4}$
(C) $x^2y - \sin(x) = \frac{\pi^2}{4} - 1$
(D) $x^2y + \sin(x) = \frac{\pi^2}{4} + 1$

Question 96 (JAM 2014 • Q14 • ID: JAM_2014_Q14) MCQ +2 M Neg: −0.67

Q.14 Let $y_1(x)$ and $y_2(x)$ be two linearly independent solutions of the differential equation

$$x^2y''(x) - 2xy'(x) - 4y(x) = 0 \text{ for } x \in [1, 10].$$

Consider the Wronskian $W(x) = y_1(x)y_2'(x) - y_2(x)y_1'(x)$. If $W(1) = 1$, then $W(3) - W(2)$ equals

(A) 1 (B) 2 (C) 3 (D) 5

Question 97 (JAM 2014 • Q4 • ID: JAM_2014_Q4) MCQ +1 M Neg: −0.33

Q.4 For $a, b, c \in \mathbb{R}$, if the differential equation

$$\left(ax^2+bxy+y^2\right)dx+\left(2x^2+cxy+y^2\right)dy=0$$

is exact, then

(A) $b=2,\ c=2a$ (B) $b=4,\ c=2$ (C) $b=2,\ c=4$ (D) $b=2,\ a=2c$

Question 98 (JAM 2012 • Q6 • ID: JAM_2012_Q6) MCQ +6 M Neg: −2.0

Q.6 An integrating factor for the differential equation $\left(2xy+3x^2y+6y^3\right)dx+\left(x^2+6y^2\right)dy=0$ is

(A) x^3 (B) y^3 (C) e^{3x} (D) e^{3y}

Question 99 (JAM 2012 • Q5 • ID: JAM_2012_Q5) MCQ +6 M Neg: −2.0

Q.5 The differential equation $\left(1+x^2y^3+\alpha x^2y^2\right)dx+\left(2+x^3y^2+x^3y\right)dy=0$ is exact if α equals

(A) $\frac{1}{2}$ (B) $\frac{3}{2}$ (C) 2 (D) 3

Question 100 (JAM 2011 • Q6 • ID: JAM_2011_Q6) MCQ +6 M Neg: −2.0

Q.6 If y^a is an integrating factor of the differential equation $2xy\,dx - (3x^2 - y^2)\,dy = 0$, then the value of a is

(A) −4 (B) 4 (C) −1 (D) 1

Question 101 (JAM 2011 • Q5 • ID: JAM_2011_Q5) MCQ +6 M Neg: −2.0

Q.5 The solution $y(x)$ of the differential equation $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = 0$ satisfying the conditions $y(0) = 4,\ \frac{dy}{dx}(0) = 8$ is

(A) $4e^{2x}$ (B) $(16x+4)e^{-2x}$ (C) $4e^{-2x} + 16x$ (D) $4e^{-2x} + 16xe^{2x}$

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Question 102 (JAM 2010 • Q10 • ID: JAM_2010_Q10) MCQ +6 M Neg: −2.0

Q.10 Consider the differential equation $(x+y+1)\,dx + (2x+2y+1)\,dy = 0$. Which of the following statements is true?

(A) The differential equation is linear
(B) The differential equation is exact
(C) e^{x+y} is an integrating factor of the differential equation
(D) A suitable substitution transforms the differentiable equation to the variables separable form

Question 103 (JAM 2010 • Q9 • ID: JAM_2010_Q9) MCQ +6 M Neg: −2.0

Q.9 Consider the differential equation $\frac{dy}{dx} = ay - by^2$, where $a, b > 0$ and $y(0) = y_0$. As $x \to +\infty$, the solution $y(x)$ tends to

(A) 0 (B) $\frac{a}{b}$ (C) $\frac{b}{a}$ (D) y_0

Question 104 (JAM 2009 • Q14 • ID: JAM_2009_Q14) MCQ +6 M Neg: −2.0

Q.14 Let $f, g:[-1, 1]\to\mathbb{R}$, $f(x)=x^3$, $g(x)=x^2|x|$. Then

(A) f and g are linearly independent on $[-1,1]$
(B) f and g are linearly dependent on $[-1,1]$
(C) $f(x)g'(x)-f'(x)g(x)$ is NOT identically zero on $[-1,1]$
(D) there exist continuous functions $p(x)$ and $q(x)$ such that f and g satisfy $y''+py'+qy=0$ on $[-1,1]$

Question 105 (JAM 2009 • Q12 • ID: JAM_2009_Q12) MCQ +6 M Neg: −2.0

Q.12 Consider the differential equation $2\cos(y^2)\,dx-xy\sin(y^2)\,dy=0$. Then

(A) e^x is an integrating factor
(B) e^{-x} is an integrating factor
(C) $3x$ is an integrating factor
(D) x^3 is an integrating factor

Question 106 (JAM 2008 • Q14 • ID: JAM_2008_Q14) MCQ +6 M Neg: −2.0

Q.14 Let $y_1(x)$ and $y_2(x)$ be twice differentiable functions on an interval I satisfying the differential equations $\frac{dy_1}{dx}-y_1-y_2=e^x$ and $2\frac{dy_1}{dx}+\frac{dy_2}{dx}-6y_1=0$. Then $y_1(x)$ is

(A) $C_1e^{-2x}+C_2e^{3x}-\frac{1}{4}e^x$
(B) $C_1e^{2x}+C_2e^{-3x}+\frac{1}{4}e^x$
(C) $C_1e^{2x}+C_2e^{-3x}-\frac{1}{4}e^x$
(D) $C_1e^{-2x}+C_2e^{3x}+\frac{1}{4}e^x$

Question 107 (JAM 2008 • Q9 • ID: JAM_2008_Q9) MCQ +6 M Neg: −2.0

Q.9 Let $y_1(x)$ and $y_2(x)$ be linearly independent solutions of the differential equation $y''+P(x)y'+Q(x)y=0$, where $P(x)$ and $Q(x)$ are continuous functions on an interval I. Then $y_3(x)=a\,y_1(x)+b\,y_2(x)$ and $y_4(x)=c\,y_1(x)+d\,y_2(x)$ are linearly independent solutions of the given differential equation if

(A) $ad=bc$ (B) $ac=bd$ (C) $ad\neq bc$ (D) $ac\neq bd$

Question 108 (JAM 2007 • Q14 • ID: JAM_2007_Q14) MCQ +6 M Neg: −2.0

14. One of the integrating factors of the differential equation $(y^2-3xy)dx+(x^2-xy)dy=0$ is

(A) $1/(x^2y^2)$
(B) $1/(x^2y)$
(C) $1/(xy^2)$
(D) $1/(xy)$

Question 109 (JAM 2007 • Q2 • ID: JAM_2007_Q2) MCQ +6 M Neg: −2.0

2. If k is a constant such that $xy+k=e^{(x-1)^2/2}$ satisfies the differential equation $x\frac{dy}{dx}=(x^2-x-1)y+(x-1)$, then k is equal to

(A) 1
(B) 0
(C) −1
(D) −2

Question 110 (JAM 2006 • Q12 • ID: JAM_2006_Q12) MCQ +6 M Neg: −2.0

12. The differential equation representing the family of circles touching y-axis at the origin is

(A) linear and of first order
(B) linear and of second order
(C) nonlinear and of first order
(D) nonlinear and of second order

Question 111 (JAM 2006 • Q10 • ID: JAM_2006_Q10) MCQ +6 M Neg: −2.0

10. If $(c_1 + c_2 \ln x)/x$ is the general solution of the differential equation

$$x^2 \frac{d^2y}{dx^2} + kx\frac{dy}{dx} + y = 0, \quad x > 0,$$

then k equals

(A) 3
(B) −3
(C) 2
(D) −1

Question 112 (JAM 2005 • Q3 • ID: JAM_2005_Q3) MCQ +6 M Neg: −2.0

3. The general solution of $x^2 \frac{d^2y}{dx^2} - 5x\frac{dy}{dx} + 9y = 0$ is

(A) $(c_1 + c_2 x)e^{3x}$

(B) $(c_1 + c_2 \ln x)x^3$

(C) $(c_1 + c_2 x)x^3$

(D) $(c_1 + c_2 \ln x)e^{x^3}$

(Here c_1 and c_2 are arbitrary constants.)

Question 113 (JAM 2005 • Q2 • ID: JAM_2005_Q2) MCQ +6 M Neg: −2.0

2. An integrating factor of $x\frac{dy}{dx} + (3x+1)y = xe^{-2x}$ is

(A) $x\,e^{3x}$

(B) $3x\,e^{x}$

(C) $x\,e^{x}$

(D) $x^3 e^{x}$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Differential Equations (2005–2026 Archive) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	NAT	+2	−1.0 to −1.0
2	NAT	+1	8.0 to 8.0
3	MSQ	+2	B;D
4	MCQ	+2	B
5	MCQ	+2	D
6	MCQ	+2	C
7	MCQ	+1	D
8	NAT	+2	3.98 to 4.02
9	NAT	+2	0.98 to 1.02
10	NAT	+1	1.39 to 1.43
11	MSQ	+2	A;B;D
12	MSQ	+2	A;B;C
13	MCQ	+2	A
14	MCQ	+2	C
15	MCQ	+2	B
16	MCQ	+2	B
17	MCQ	+1	A
18	MCQ	+1	B
19	NAT	+2	6 to 6
20	NAT	+1	1.0 to 1.0
21	NAT	+1	0.19 to 0.21
22	MCQ	+2	B
23	MCQ	+2	A
24	MCQ	+2	A
25	MCQ	+1	B
26	MCQ	+1	A
27	MCQ	+1	A
28	MCQ	+1	B
29	NAT	+2	−0.01 to 0.01
30	NAT	+2	0.49 to 0.51
31	NAT	+1	−0.01 to 0.01
32	MSQ	+2	B, D
33	MCQ	+2	A
34	MCQ	+1	D
35	NAT	+1	2.00 to 2.00
36	NAT	+1	2.24 to 2.26
39	MSQ	+2	C,D
40	MSQ	+2	A,C
41	MCQ	+2	B
42	MCQ	+2	A
43	MCQ	+2	D
44	MCQ	+2	B
45	NAT	+2	10.8 to 10.9
46	NAT	+1	3
47	MCQ	+2	B
48	MCQ	+2	A
49	MCQ	+2	D
50	MCQ	+1	A
51	MCQ	+1	D
52	NAT	+2	3 to 3
53	NAT	+2	8 to 8
54	NAT	+1	7 to 7
55	NAT	+1	0.1 to 0.1
56	MSQ	+2	A;C
57	MCQ	+2	A
58	MCQ	+2	C
59	MCQ	+1	D
60	NAT	+2	1.12 to 1.25
61	NAT	+2	4 to 4
62	MCQ	+2	D
63	MCQ	+2	C
64	MCQ	+1	C
65	MCQ	+1	A
66	NAT	+2	0.5 to 0.5
67	NAT	+2	−2.80 to −2.70
68	NAT	+1	9 to 9
69	MSQ	+2	A;B;C
70	MCQ	+2	B
71	MCQ	+2	C
72	MCQ	+2	A
73	MCQ	+1	B
74	NAT	+2	1.34 to 1.36
75	NAT	+2	2.9 to 3.1
76	NAT	+1	1.9 to 2.1
77	MSQ	+2	C, D
78	MSQ	+2	B, C, D
79	MCQ	+2	B
80	MCQ	+2	C
81	NAT	+2	−0.3 to −0.25
82	NAT	+1	6.0 to 6.0
83	MSQ	+2	C;D
84	MCQ	+2	C
85	MCQ	+2	B
86	MCQ	+1	C
87	MCQ	+1	D
88	MSQ	+2	A;C
89	MCQ	+2	B
90	MCQ	+2	D
91	MCQ	+1	B
92	MCQ	+1	C
93	MCQ	+1	C
94	MCQ	+2	D
95	MCQ	+2	B
96	MCQ	+2	D
97	MCQ	+1	B
98	MCQ	+6	C
99	MCQ	+6	B
100	MCQ	+6	A
101	MCQ	+6	B
102	MCQ	+6	D
103	MCQ	+6	B
104	MCQ	+6	A
105	MCQ	+6	D
106	MCQ	+6	C
107	MCQ	+6	C
108	MCQ	+6	B
109	MCQ	+6	A
110	MCQ	+6	C
111	MCQ	+6	A
112	MCQ	+6	B
113	MCQ	+6	A

Q. No.	Type	Marks	Official Key
37	NAT	+1	**0 to 0**
38	NAT	+1	**0.32 to 0.34**

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, −2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.